

100 Days of Azure DevOps

Day 30 Handout — Same spine in every language: install, build, test, publish a drop — secrets stay in the group

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — Phase 3 map (Days 21–29)

Day	Object you should still be able to draw	Keep
21–22	pool: vmlImage vs named self-hosted pool	Hosted ubuntu-latest until a private-network constraint
23 + 28	trigger/pr → stages → jobs → steps; dependsOn; matrix	Sketch from memory or you copied
24–27	.NET / Node / Python / Maven CI tasks	One primary stack green; others may stay stubs
29	Library group + secret mask	No passwords in azure-pipelines.yml
Handoff	PublishPipelineArtifact@1 name: drop	CD (Phase 4) must download this, not rebuild

Architecture B — pick depth, not four half-pipelines

Choice	Definition of done today	Trap
.NET primary	restore/build/test/publish + drop on main	Keeping Node/Python red 'for later'
Node primary	NodeTool + npm ci + test + drop	Skipping lockfile so CI ≠ laptop
Python primary	3.11 + pytest collected ≥1 + drop (even a wheel/zip)	Lint true with no note
Java primary	JDK 17 + mvn -B test/package + drop	-DskipTests to force a screenshot

Architecture C — recap failures that fake a phase close

Pitfall	What you see	Fix
No artifact	Green job, nothing to promote Monday	PublishPipelineArtifact@1 before you write the recap
Secret in shot	Library value visible in a LinkedIn PNG	Retake. Mask or crop. Rotate the dummy if leaked
Copied YAML only	Cannot list trigger → stage → job → step	Draw it once in docs/phase3-recap.md
Four languages, zero green	A graveyard of samples	One green main. Stubs are honest; red mains are not

One-liner to remember

Phase 3 recap: CI is a seatbelt you wear before the crash, not after — and the buckle is a drop you can promote.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org only. Project azure-100-labs. Pick one primary stack. Personal time, personal repo.

- 1 Choose primary stack (.NET Day 24, Node 25, Python 26, or Maven 27). Delete or ignore the other pipelines if they distract — do not keep red mains.
- 2 Ensure that pipeline publishes an artifact named drop (PublishPipelineArtifact@1 or PublishBuildArtifacts@1). Run on main until green.
- 3 Open the run → Artifacts and confirm drop exists. That object is the Phase 4 input.
- 4 Write docs/phase3-recap.md: (1) YAML anatomy in one line, (2) one war story from this week (cache miss, skipped test, almost-echoed secret).

- 5 Screenshot the green run with no variable values. Save it locally — do not commit secrets or org URLs you would not post.
- 6 Pipelines folder is the source of truth. If Classic UI still holds a leftover, leave it unused.

Artifact handoff — publish drop (adapt path to your stack)

```
# append to your primary CI job
- task: PublishPipelineArtifact@1
  inputs:
    targetPath: $(Build.ArtifactStagingDirectory)
    artifact: drop
    publishLocation: pipeline
# Phase 4 will: download: current artifact: drop
# Do not dotnet/npm/mvn publish again in a release job.
```

Done checklist

- One green CI on main for the primary stack
- Artifact drop is visible on that run
- docs/phase3-recap.md has anatomy + one war story
- No secrets in YAML or screenshots

Tomorrow — Day 31

Release pipelines overview — Classic is literacy; YAML CD is a deployment job + environment: dev.